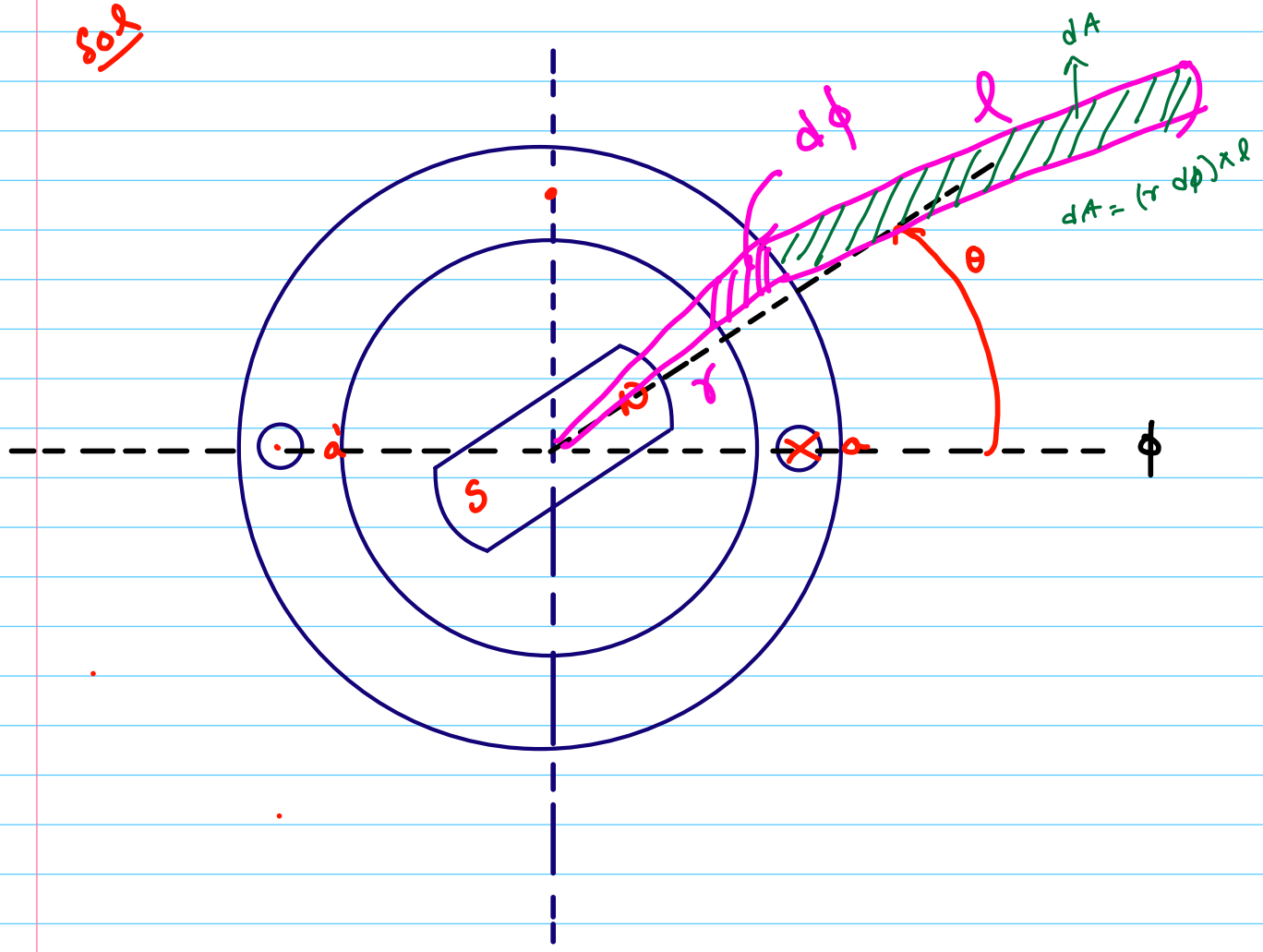
EE2005 : TUTORIAL - 8 on Induction Machines

1. Consider a stator coil of N turns, with the sides of the coil separated 180° apart. The length of the coil side is l. The coil links a flux density waveform produced by the rotor which varies as $B(\phi) = B_m \cos(\phi)$, where ϕ is the angle around the circumference of the stator. With respect to this waveform, the coil sides may be taken to be at angles $\phi = 0°$ and $\phi = 180°$ and the rotor field axis positioned at $\phi = 0^o$. With respect to this waveform, the coil sides may be taken to be at angles $\phi = 0^o$ and $\phi = 180^o$.

   (a) If this flux density waveform now rotates in the air gap due to the rotation of the rotor, find the expression for B when the rotor is at some angle $\theta = \omega t$.

   (b) Derive an expression for the flux linkage of the stator coil at any angle θ of the rotor. (Hint: Find the flux passing through an elemental area at an angle ϕ, and integrate to find the total flux passing through the area spanned by the coil).

Sol



(a)

Initially rotor axis is along $\phi = 0$.

Air gap flux density

$$B(\phi) = B_m \cos\phi$$

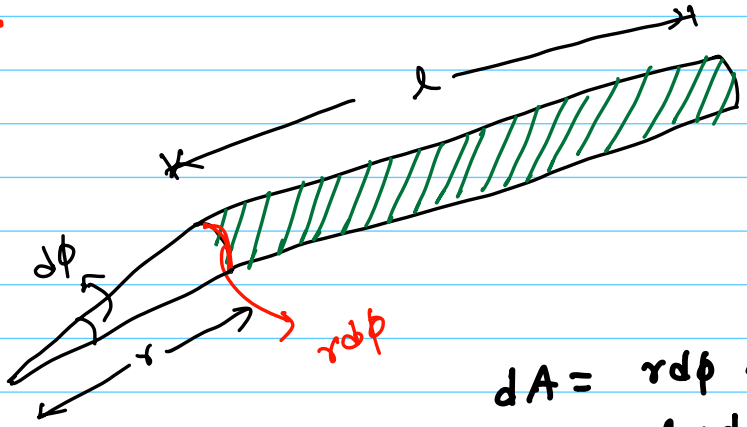
Rotor is moved through an angle $\theta = \omega t$

Now, Air gap flux density

$$B(\phi, \theta) = B_m \cos(\phi - \theta)$$

(b)

The differential area element be dA on stator surface.



$$dA = r\,d\phi\, l$$
$$= l r\, d\phi$$

The flux passing through dA is $d\psi$

$$B = \frac{d\psi}{dA}$$

$$d\psi = B\, dA$$

$$d\varphi = B_m \cos(\phi - \theta)\, l r\, d\phi$$

Net flux passing through the coil = φ

$$= \int_0^{\pi} d\varphi$$

$$= \int_0^{\pi} B_m \cos(\phi - \theta)\, l r\, d\phi$$

$$\varphi = B_m l r \int_0^{\pi} \cos(\phi - \theta)\, d\phi$$

$$= B_m l r \left[\sin(\phi - \theta)\right]_0^{\pi}$$

$$= B_m l r\, 2 \sin\theta$$

$$\boxed{\varphi = 2 B_m l r \sin\theta}$$

Total flux linkages $\lambda = N\varphi$

$$= 2 N B_m l r \sin\theta$$

EMF induced $e = \dfrac{d\lambda}{dt}$

$$= \frac{2 N B_m l r \sin \omega t}{dt}$$

$$e = 2 N B_m l r\, \omega \cos \omega t$$

2. Now consider another identical coil placed in the adjacent slot, which is situated at an angle of δ away from the first, i.e the coil sides may be assumed to lie at angles $\phi = \delta$ and $\phi = \pi + \delta$. Derive the expression for the voltage induced in that coil. If the two coils are now connected in series, what would be the net voltage?

Sol

Total flux linkages through coil 1

$$\lambda_1 = N\phi_1$$
$$= 2NB_m l r \sin\theta$$

EMF induced in coil 1

$$e_1 = \frac{d\lambda_1}{dt}$$
$$= \frac{2NB_m l r \sin\omega t}{dt}$$
$$e_1 = 2NB_m l r \omega \cos\omega t$$

An adjcent slot at angle δ and $\pi+\delta$ with respect to coil 1 and coil 2 is placed

Now the flux linkages through coil 2

$$\psi_2 = N B_m r l \int_{\delta}^{\pi+\delta} \cos(\phi - \theta)\, d\phi$$

$$= N B_m r l \left(2 \sin(\theta - \delta)\right)$$

$$\psi_2 = 2 N B_m r l \sin(\theta - \delta)$$

EMF induced in coil 2

$$e_2 = \frac{d\psi_2}{dt}$$

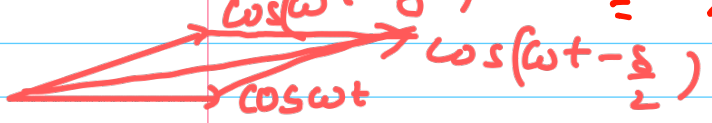
$$= 2 N B_m r l \omega \cos(\theta - \delta)$$

$$e_2 = 2 N B_m r l \omega \cos(\omega t - \delta)$$

two coil are connected in series the resultant emf

$$e = e_1 + e_2$$

$$= 2 N B_m r l \omega \cos \omega t + 2 N B_m r l \omega \cos(\omega t - \delta)$$



$$= 2 N B_m r l \omega \left[\cos \omega t + \cos(\omega t - \delta)\right]$$

$$e = 4 N B_m r l \omega \cos\frac{\delta}{2} \cos\left(\omega t - \delta/2\right)$$

3. A 208 V, 10 hp, 4 pole, 60 Hz, Y-connected induction motor has a full-load slip of 5 %.

(a) What is the synchronous speed of this motor?

(b) What is the rotor speed of this motor at the rated load?

(c) What is the rotor frequency of this motor at the rated load?

(d) What is the shaft torque of this motor at the rated load?

Sol

(a) $N_S = \frac{120f}{P} = \frac{120 \times 60}{4} = 1800$ RPM

(b) $N_r =$ ———

$$s = \frac{N_S - N_r}{N_S} \Rightarrow N_r = (1-s)N_S$$

Full load slip

$$s_{full} = 0.05$$

$$N_r = (1 - s_{full}) N_S$$

$$= (1 - 0.05)\ 1800$$

$$N_r = 1710 \text{ RPM}$$

(c) $f_r = sf$

$$f_r = s_{full} f$$

$$= 0.05 \times 60$$

$$f_r = 3 \text{ Hz}$$

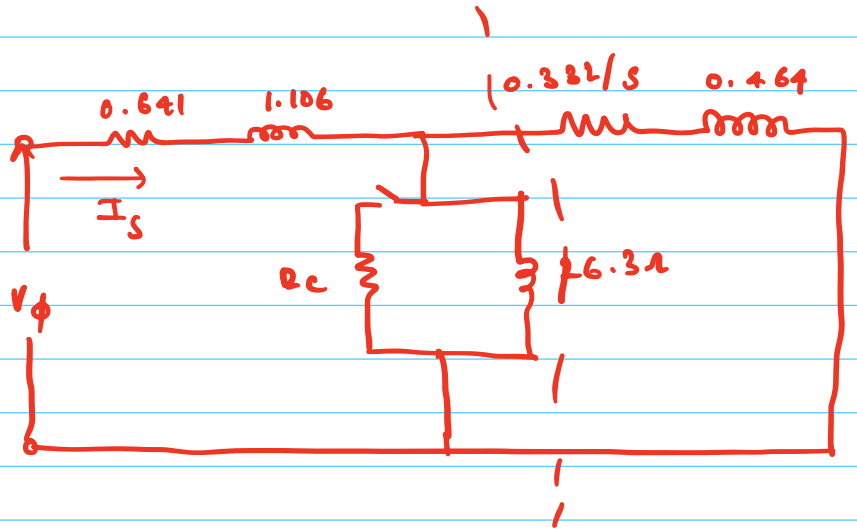
(d) $T_{shaft} = \dfrac{P_{output\ (or)}\ P_{shaft}}{\omega_r}$

$$= \frac{10 \times 745.7}{2\pi \frac{N_r}{60}}$$

$T_{shaft} = 41.6427$ N-m

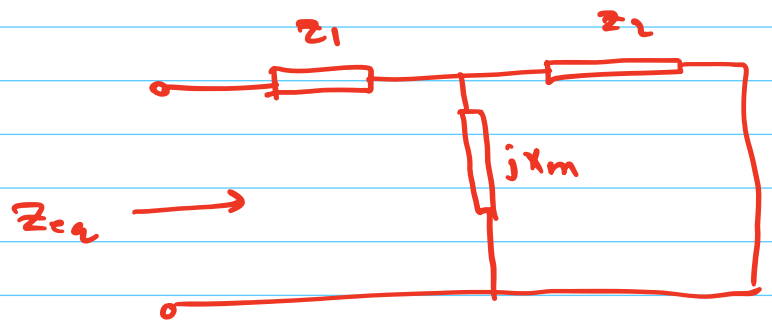
4. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
$R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. For a rotor slip of 2.2% at the rated voltage and rated frequency, find the motor's

   (a) Speed
   (b) Stator current
   (c) Power factor
   (d) Resistance equivalent to mechanical load



$$N_s = \frac{120f}{P} = \frac{120 \times 60}{4} = 1800 \text{ RPM}$$

(a) $N_r = (1-s)N_s = (1-0.02)\,1800 = 1760.4$ RPM



$$Z_{eq} = Z_1 + Z_2 \,//\, jX_m$$

$$= Z_1 + \frac{jX_m Z_2}{Z_2 + jX_m}$$

$$Z_{eq} = 14.0579 \angle 33.6826° \ \Omega$$

$$I_s = \frac{V_\phi}{Z_{eq}} = \frac{460/\sqrt{3} \angle 0°}{14.0579 \angle 33.6826}$$

$$I_s = 18.891 \angle -33.6826° \ A$$

(c) $pf_1 = \cos\phi_1 = \cos(-33.6826°)$

$$pf_1 = 0.8321 \text{ lag}$$

(d) Resistance equivalent to Mech. load $= R_2' \left(\frac{1-s}{s}\right)$

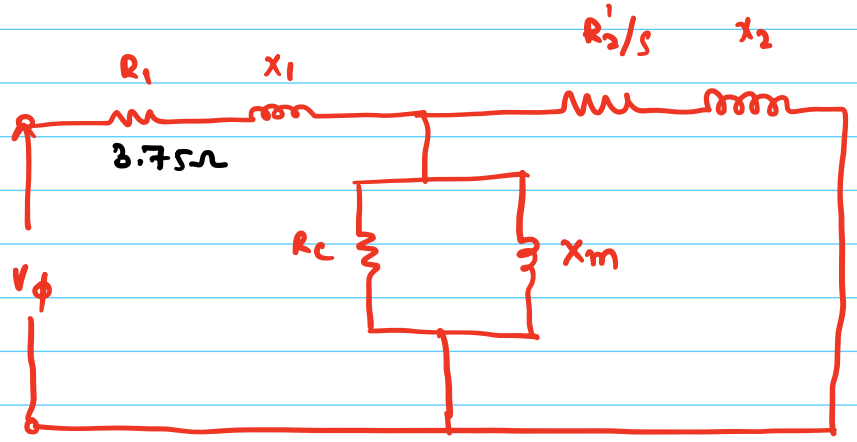
$$= 0.0332 \times \frac{1-0.022}{0.022}$$

$$= 14.759 \ \Omega$$

5. The following results were obtained on a three phase, 75 kW, 3.3 kV, 6-pole, 50 Hz, Y-connected squirrel cage induction motor.
No load Test: $f = 50\,\text{Hz}$, $V_0 = 3.3\,\text{kV}$, $I_0 = 5\,\text{A}$, $P_0 = 2500\,\text{W}$

Blocked rotor Test: $f = 15\,\text{Hz}$, $V_{BR} = 400\,\text{V}$, $I_{BR} = 27\,\text{A}$, $P_{BR} = 15\,\text{kW}$
The stator resistance per phase is $3.75\,\Omega$.

Sol



$$V_\phi = \frac{V}{\sqrt{3}}$$

To find R_c & X_m:

Given $V_0 = 3.3\,\text{kV}$

$$V_{\phi 0} = \frac{V_0}{\sqrt{3}} = \frac{3.3\,\text{k}}{\sqrt{3}}$$

$$I_0 = 5\,\text{A}$$

$$P_0 = 2500\,\text{W}$$

$$P_0 = P_{\text{irom}} + P_{\text{mech}} + P_{\text{stator loss}}$$

$$= P_{\text{irom}} + P_{\text{mech}} + 3 I_0^2 R_1$$

P_{mech} are given so assume $P_{mech} = 0$

$\therefore \quad P_0 = P_{iron} + 3 I_0^2 R_1$

$P_{iron} = P_0 - 3 I_0^2 R_1 = 2500 - 3 \times 5^2 \times 3.75$

$P_{iron} = 2218.75 \text{ W}$

$P_{iron} = \sqrt{3}\, V_0 I_0 \cos\phi_0$

$\cos\phi_0 = \dfrac{P_{iron}}{\sqrt{3}\, V_0 I_0}$

$= \dfrac{2218.75}{\sqrt{3} \times 3.3 \times 1000 \times 5}$

$\cos\phi_0 = 0.0776 \text{ lag}$

$\phi_0 = \cos^{-1}(0.0776)$

$\phi_0 = 85.55^\circ$

$I_\mu = I_0 \sin\phi_0 = 5 \times \sin(85.55)$

$I_\mu = 4.985 \text{ A}$

$$I_w = I_0 \cos\phi_0 = 5 \times \cos(85.55)$$

$$I_w = 0.388\ A$$

$$R_c = \frac{V_{\phi 0}}{I_w} = \frac{3.3 \times 1000/\sqrt{3}}{0.388}$$

$$R_c = 4908.169\ \Omega/ph$$

$$X_m = \frac{V_{\phi 0}}{I_\mu} = \frac{3.3 \times 1000/\sqrt{3}}{4.985}$$

$$X_m = 382.204\ \Omega/ph$$

→

$R_1 = 3.75\ \Omega/ph$

$R_c = 4908.169\ \Omega/ph$

$X_m = 382.204\ \Omega/ph$

To find X_1, X_2' & R_2'

$$Z_{BR} = \frac{V_{\phi BR}}{I_{BR}}$$

$$= \frac{400/\sqrt{3}}{27}$$

$$Z_{BR} = 8.553\ \Omega$$

$$R_{BR} = R_1 + R_2' = \frac{P_{BR}}{3 I_{BR}^2}$$

$$= \frac{15 \times 10^3}{3 \times (27)^2}$$

$$R_1 + R_2' = 6.8587\ \Omega$$

$$R_2' = 6.8587 - 3.75$$

$$R_2' = 3.1011\ \Omega$$

$$X_{BR} = X_1 + X_2'$$

$$X_{BR} = \sqrt{Z_{BR}^2 - R_{BR}^2}$$

$$= \sqrt{8.553^2 - 6.8587^2}$$

$$X_{BR} = 5.1105\ \Omega$$

$$X_1 = X_1'$$

$$\Rightarrow \quad 2X_1 = X_{BR}$$

$$X_1 = \frac{X_{BR}}{2}$$

But X_{BR} we got at $f = 15\,kHz$

we need at $50\,kHz$

so, as we know

$$X \propto f$$

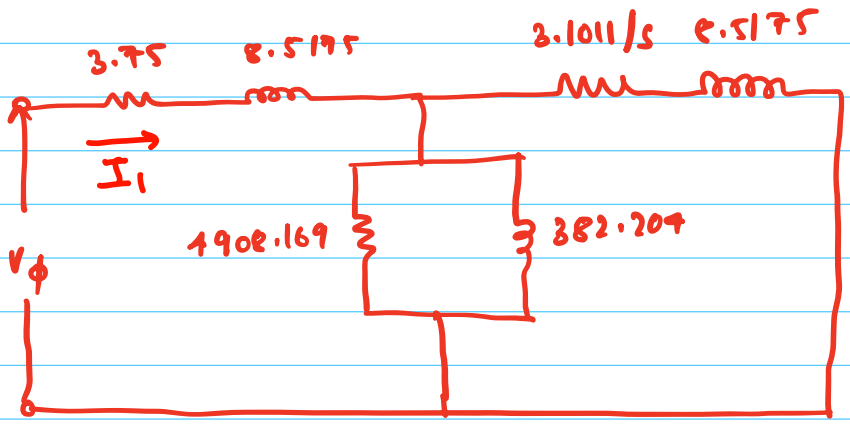
$$\Rightarrow \frac{X_{BR@50}}{X_{BR@15}} = \frac{50}{15}$$

$$X_{BR@50} = X_{BR@15} \times \frac{50}{15}$$

$$= 5.1105 \times \frac{50}{15}$$

$$X_{BR@50} = 17.035$$

$$X_1 = \frac{X_{BR@50}}{2} = \frac{17.035}{2} = 8.5175\,\Omega = X_1'$$



(b)

$s = 0.04$

$$Z_3 = \frac{3.1011}{0.04} + j\,8.5175$$

$$= 77.7177 + j\,8.5175$$

$$Z_2 = 4908.169 \;||\; j382.204$$

$$= \frac{4908.169 \times j382.204}{4908.169 + j382.204}$$

$$= 29.583 + j\,379.901$$

$$Z_4 = Z_2 \;||\; Z_3$$

$$= (29.583 + j379.901) \;||\; (77.7177 + j8.5175)$$

$$= 70.607 + j\,21.916$$

$$Z_{eff} = (R_1 + jX_1) + Z_4$$

$$= 3.75 + j8.5175 + 70.607 + j21.916$$

$$= 74.357 + j30.434$$

$$= 80.345 \angle 22.259 \ \Omega$$

$$I_1 = \frac{V_\phi}{Z_{eff}}$$

$$= \frac{3.3 \times 10^3 / \sqrt{3}}{80.345 \angle 22.259}$$

$$I_1 = 23.713 \angle -22.259 \ A$$

$$\text{primary pf} = \cos(22.259)$$

$$= 0.925 \ \text{lag}$$

$$I_2' = I_1 \times \frac{Z_3}{Z_2 + Z_3}$$

$$= 23.713 \angle -22.259 \times \frac{29.583 + j379.901}{77.717 + j8.517 + 29.583 + j379.901}$$

$$= 22.423 \angle -11.269 \ A$$

Total losses = $P_{SCL} + P_{iron} + P_{RCL}$

$= 3|I_1|^2 R_1 + 2218.75 + 3|I_2'|^2 R_2$

$= 3 \times (23.713)^2 \times 3.75$

$+ 2218.75$

$+ 3 \times (22.423)^2 \times 3.1087)$

$= 13233.791\ W$

$P_{in} = \sqrt{3}\, V_L I_1 \cos\phi$

$= \sqrt{3} \times 3.3 \times 10^3 \times 23.713 \times 0.925$

$= 125372.6487\ W$

$\% \eta = \frac{P_o}{P_i} \times 100$

$= \frac{P_i - \text{losses}}{P_i} \times 100$

$= \frac{125372.6487 - 13233.791}{125372.6487} \times 100$

$= 89.44\ \%$